

Preference-Calibrated Human-in-the-Loop Reinforcement Learning for Robotic Manipulation

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Abstract: Human-in-the-loop reinforcement learning (HIL-RL) improves sample efficiency in real-robot manipulation through online human intervention. However, successful trajectories may include suboptimal actions that deviate from the desired task-execution path and force human intervention. Existing HIL-RL methods typically apply the consistent credit assignment principle to all transitions, uniformly propagating discounted terminal rewards through suboptimal segments, ignoring the actual contribution of each transition to task success. This overestimates Q-values for critic learning and indirectly misguides actor updates toward suboptimal behavior patterns. To this end, we propose PACT, a **P**reference-calibrated **A**ctor-**C**ritic **T**raining framework for HIL-RL. It leverages intervention-induced preferences to reassign credit on identified suboptimal segments and guide policy optimization while mitigating Q-value overestimation. Specifically, we first design a progress model that learns from human demonstration and identifies suboptimal segments for credit correction. Then, from the human action and resampled policy action at the intervention state, we build preference pairs to define a counterfactual advantage that penalizes Bellman targets of the identified suboptimal segment, enabling directional credit calibration. Moreover, we directly align the policy with human corrective actions in the bounded mean space, providing an additional signal beyond critic-guided updates. Across six real-robot manipulation tasks, PACT improves the average success rate by 27.2% and achieves $1.3\times$ faster convergence, improving both RL sample efficiency and performance. Videos and code are available at [Project Page](#).

Keywords: Robotic Manipulation, Reinforcement Learning

1 Introduction

Deep reinforcement learning (DRL) has made substantial progress in robotic manipulation by enabling agents to acquire complex behaviors through trial-and-error interaction [1, 2]. However, simulation-trained policies often suffer from domain-transfer failures [3], while real-world exploration is constrained by high interaction cost and safety risks [4]. Human-in-the-loop RL (HIL-RL) addresses these challenges by incorporating real-time operator intervention, achieving effective policy learning under limited interaction budgets and substantially improving the sample efficiency of real-robot RL [5, 6, 7, 8].

Despite recent progress, existing HIL-RL methods remain sample-inefficient and require substantial human intervention during training. We argue that this limitation partly stems from their neglect of the intrinsic heterogeneity of intervention-containing successful trajectories, where normal exploration, suboptimal segments, and human recovery coexist. In such trajectories, task success does not imply that all actions are beneficial, since some policy actions may deviate from the desired task-

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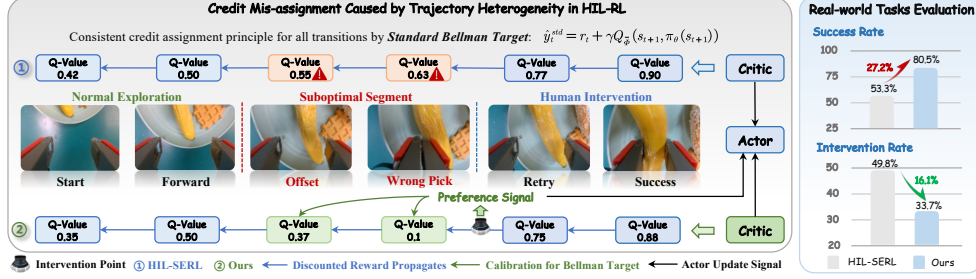


Figure 1: PACT calibrates credit assignment in intervention trajectories by localizing suboptimal segments and correcting inflated Q-values.

execution path and force human intervention. However, current methods typically apply the same credit assignment principle to all transitions, treating them as homogeneous learning samples. Under sparse terminal rewards, this uniformly propagates the final success signal through suboptimal segments, overestimating their actual contribution to task success. The resulting inflated Q-values can further misguide actor updates, since the policy is optimized to maximize the critic estimate, thereby reinforcing similar suboptimal behavior patterns and slowing convergence.

To address this issue, we propose PACT, a **P**reference-calibrated **A**ctor-**C**ritic **T**raining framework for HIL-RL. Different from existing HIL-RL methods, PACT treats intervention-containing successful trajectories as heterogeneous learning samples, performs fine-grained credit calibration by identifying suboptimal action segments according to their contribution to task success, and guides policy and preference alignment. As shown in Figure 1, PACT constructs zero-cost preference pairs from human corrective actions and policy actions at the intervention state. These preferences are used to directionally penalize the Bellman targets of identified suboptimal segments, suppressing Q-value overestimation on the critic side, while directly guiding the actor toward human-preferred corrective actions beyond conventional critic-guided updates. Across six real-robot manipulation tasks, PACT improves the average success rate from 53.3% to 80.5%, reduces the intervention rate from 49.8% to 33.7%, and achieves $1.3\times$ faster training than HIL-SERL, with no inference-time overhead. Our main contributions are summarized as follows:

- We propose PACT, a plug-and-play preference-calibrated actor-critic training framework that addresses credit misassignment in HIL-RL caused by trajectory heterogeneity.
- We design a progress model trained from human demonstrations to locate suboptimal segments for critic correction, and further build preference pairs to perform both critic- and actor-side calibration, including credit reassignment and policy optimization guidance.
- We conduct extensive experiments on six real-robot manipulation tasks, demonstrating improved success rate, reduced human intervention, and faster convergence.

2 Related Work

2.1 Human in the loop RL for Robotic Manipulation

Real-robot RL learns policies through direct interaction with physical environments, but its sample efficiency is severely constrained by costly data collection and safety risks, especially compared with large-scale parallel simulation [9, 10, 11]. Prior work improves real-world efficiency through model-based RL [12, 13], hybrid approaches [9, 14], and priors from pretrained foundation models [15, 16], but these methods often introduce additional modeling, training, or deployment overhead. Human-in-the-loop RL offers a practical alternative by allowing operators to intervene during online exploration. Early human-robot collaborative learning builds on interactive imitation learning [17, 18], but purely imitative objectives cannot continuously improve through trial-and-error interaction. Recent methods therefore combine offline demonstrations with online RL [19, 20]. Among them, SERL [8]

provides a complete real-robot RL framework, and HIL-SERL [5] further incorporates real-time intervention trajectories into replay-based training, achieving strong performance on dexterous manipulation tasks. Nevertheless, existing HIL-RL methods still optimize all transitions uniformly, overlooking the heterogeneity in intervention-containing successful trajectories and thus inducing credit misassignment. In this paper, we address this by exploiting intervention-induced preferences for credit reassignment on identified suboptimal segments.

2.2 Q-value Correction and Preference Learning

Q-value overestimation is a long-standing challenge in reinforcement learning. Existing methods mainly address it by improving critic estimation, such as Double DQN [21], TD3 [22], and CQL [23], or by reshaping the Bellman target with additional supervisory signals. The latter includes reward shaping with dense rewards [24, 25, 26], inverse RL from expert demonstrations [27], and preference-based learning from human comparisons [28, 29]. However, these methods typically rely on global trajectory-level correction, external reward models, or costly human annotation, and thus do not address segment-level credit errors inside intervention-containing trajectories. In contrast, HIL-RL naturally provides a zero-cost preference signal: at each intervention point, the human corrective action is preferred to the policy action. Based on this signal, our method performs preference-calibrated segment-level credit correction to reshape Bellman targets and further constrains actor learning for more sample-efficient policy optimization.

3 Methodology

In this section, we first introduce the problem of credit misassignment in HIL-RL (3.1). Then, we present a lightweight progress model used to localize suboptimal segments (3.2). Building on this, we propose a preference-aware counterfactual Q-value correction method to suppress overestimation (3.3). Finally, we introduce a preference auxiliary policy optimization that directly guides policy learning at intervention points (3.4). Our overall pipeline can be seen in Figure 2.

3.1 Preliminaries and Problem Statement

Real-robot manipulation can be formulated as a Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{P}(s_{t+1}|s_t, a_t)$ is the transition dynamics, $r(s_t, a_t)$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor. The objective is to learn a policy $\pi_\theta(a_t|s_t)$ that maximizes the expected cumulative discounted return $\mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right]$.

Recent HIL-RL methods improve the sample efficiency by incorporating human intervention and optimizing the policy with RLPD [20]. The critic is trained with the standard Bellman target:

$$\hat{y}_t^{std} = r_t + \gamma Q_{\bar{\phi}}(s_{t+1}, \pi_\theta(s_{t+1})), \quad \mathcal{L}_{\text{critic}} = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathcal{B}} \left[(Q_\phi(s_t, a_t) - \hat{y}_t^{std})^2 \right], \quad (1)$$

where $\mathcal{B}$ is the replay buffer, and $Q_{\bar{\phi}} \triangleq \min_{i=1,2} Q_{\phi_i}$ denotes the clipped target Q-value. The actor is updated by maximizing the critic estimate with entropy regularization:

$$\mathcal{L}_{\text{actor}} = -\mathbb{E}_{s_t \sim \mathcal{B}} [Q_\phi(s_t, \pi_\theta(s_t)) + \alpha \mathcal{H}(\pi_\theta(\cdot|s_t))], \quad (2)$$

where α is the temperature coefficient and $\mathcal{H}(\cdot)$ is the entropy loss.

However, this objective applies the same backup to all transitions. An intervention-containing successful trajectory is heterogeneous and can be written as $\tau = \tau^{\text{norm}} \circ \tau^{\text{sub}} \circ \tau^{\text{hum}}$, where $\tau^{\text{norm}} = \{s_t, a_t\}_{t=0}^{t_a-1}$, $\tau^{\text{sub}} = \{s_t, a_t\}_{t=t_a}^{t_h-1}$, and $\tau^{\text{hum}} = \{s_t, a_t\}_{t=t_h}^{T-1}$ denotes the normal, sub-optimal, and human intervention segments, respectively. Under sparse terminal rewards, Eq. (1) propagates the success signal through the entire trajectory, assigning undeserved credit to actions in τ^{sub} and resulting in Q-values overestimation. In light of this, we aim to address this issue by representing such trajectories as structured heterogeneous data and handling them at a finer granularity during training, enabling more accurate credit assignment and more efficient policy optimization.

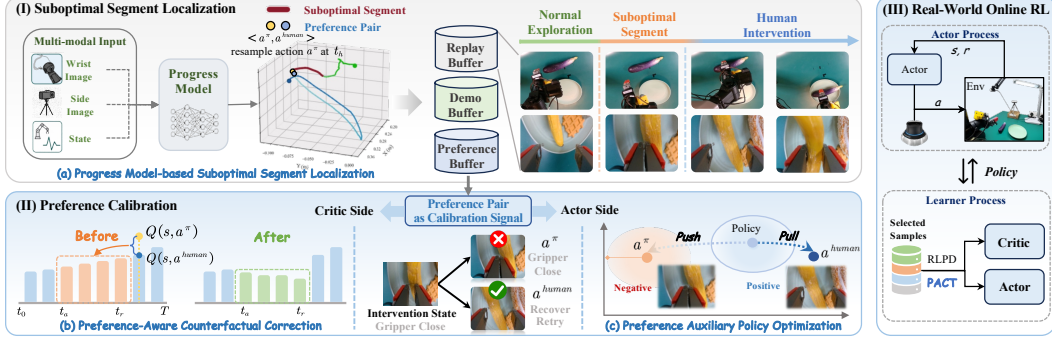


Figure 2: PACT converts intervention-containing trajectories into preference-calibrated critic and actor updates for sample-efficient real-world HIL-RL.

3.2 Lightweight Progress Model-based Suboptimal Segment Localization

To localize the suboptimal segments, we design a lightweight task progress estimator $f_\psi : \mathcal{O} \rightarrow [0, 1]$ that maps observations to a scalar progress value. It is trained in a fully self-supervised manner using only the demonstration data $\mathcal{D}_{\text{demo}}$, with *zero* additional data collection or annotation cost.

Input Representation. We employ an encoder $g_{\text{img}} : g_{\text{res}} \cdot h$ to encode image data and a learnable MLP $g_{\text{prop}} : \mathbb{R}^7 \rightarrow \mathbb{R}^{d_v}$ to encode robot proprioceptive data, where $g_{\text{res}} : \mathbb{R}^{H \times W \times 3} \rightarrow \mathbb{R}^{d_v}$ is a frozen pretrained ResNet encoder, and $h : \mathbb{R}^{d_v} \rightarrow \mathbb{R}^{d_z}$ is a learnable projection. Let I_t^w, I_t^s denote wrist and side camera images. The observation is constructed from residuals relative to the initial frame and proprioceptive displacement:

$$o_t = [g_{\text{img}}(I_t^w - I_0^w), g_{\text{img}}(I_t^s - I_0^s), g_{\text{prop}}(q_t - q_0)], \quad \hat{p}_t = f_\psi(o_t). \quad (3)$$

The progress model f_ψ then predicts a scalar value $\hat{p}_t = f_\psi(o_t) \in [0, 1]$.

Self-Supervised Training. Building on the assumption that successful demonstration trajectories are near-optimal and exhibit smooth, monotonic task progress, we use uniform progress $p_t^* = t/T$ as pseudo-labels for a trajectory of T steps. Furthermore, the regression loss with a discounted Monte Carlo return is introduced to strengthen the terminal supervision over the final segment $\mathcal{T}_{\text{end}}$:

$$\mathcal{L}_{\text{progress}} = \frac{1}{T} \sum_{t=1}^T \left(f_\psi(o_t) - \frac{t}{T} \right)^2 + \frac{\lambda_{\text{mc}}}{|\mathcal{T}_{\text{end}}|} \sum_{t \in \mathcal{T}_{\text{end}}} \left(f_\psi(o_t) - \sum_{k=0}^{T-t-1} \gamma^k r_{t+k} \right)^2. \quad (4)$$

Suboptimal Segment Localization. We perform progress inference on intervention-containing successful trajectories to localize suboptimal segments by localizing abnormal progress patterns before interventions. Let $\hat{p}_t = f_\psi(o_t)$ and $\Delta_w(t) = \hat{p}_{t+w} - \hat{p}_t$, we identify a regression anchor when the predicted progress within a short future window drops below the current progress by a margin:

$$t_a = \min \{ t < t_h \mid \exists w \in \{1, \dots, \min(W, t_h - t)\}, \hat{p}_{t+w} < \hat{p}_t - \delta_{\text{reg}} \}, \quad (5)$$

where the recovery point is defined as the first timestep after t_a with K consecutive progress increases:

$$t_r = \min \{ t > t_a \mid \hat{p}_{t+k} - \hat{p}_{t+k-1} > 0, \forall k = 1, \dots, K \}. \quad (6)$$

If no recovery is found before intervention, we set $t_r = t_h - 1$. The resulting interval $[t_a, t_r]$ is treated as a candidate suboptimal segment for subsequent critic correction.

3.3 Preference-Aware Counterfactual Correction

Given the identified suboptimal segments, PACT calibrates critic learning by converting preferences into target-level corrections. The key idea is to measure how much the current critic violates the preference that human corrective actions should be favored over suboptimal policy actions.

Counterfactual Advantage. Human intervention induces a local preference: at the intervention state, the corrective action $a_{t_h}^{\text{human}}$ should be valued above the policy action $a_{t_h}^\pi$, i.e., $Q(s_{t_h}, a_{t_h}^{\text{human}}) > Q(s_{t_h}, a_{t_h}^\pi)$ under the desired value function. To operationalize this preference, we introduce a counterfactual advantage that measures the preference-violating value gap under the current target critic. This term provides a directional calibration signal for critic learning and determines how much over-credited value should be removed from the preceding suboptimal segment:

$$A_{\text{cf}} = \text{sg} \left[\max \left(Q_{\bar{\phi}}(s_{t_h}, a_{t_h}^\pi) - Q_{\bar{\phi}}(s_{t_h}, a_{t_h}^{\text{human}}), 0 \right) \right], \quad (7)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator, and the non-negative clipping ensures directional suppression of preference-violating critic estimates.

Corrected Bellman Target. Although A_{cf} is computed from a single intervention point, the detected interval $[t_a, t_r]$ corresponds to the policy-executed segment leading to the intervention. Therefore, we use A_{cf} as a segment-level correction signal to calibrate its Bellman targets.

Considering early actions near t_a may still reflect exploratory behavior and have weaker causal evidence for intervention, whereas later actions are more strongly associated with the detected suboptimal execution. We introduce a position-aware exponential weight $\omega(t) = \exp \left(-\beta \cdot \frac{t_r - t}{t_r - t_a} \right)$, where β controls how sharply the correction increases along the segment. The exponential form provides conservative correction for early uncertain actions while concentrating stronger calibration on later high-confidence suboptimal actions.

Let $\hat{y}_t^{\text{std}} = r_t + \gamma Q_{\bar{\phi}}(s_{t+1}, \pi_\theta(s_{t+1}))$ denote the standard Bellman target. PACT penalize the Bellman target by subtracting the position-weighted counterfactual correction:

$$\hat{y}_t^{\text{corr}} = \begin{cases} \hat{y}_t^{\text{std}} - \omega(t) \cdot A_{\text{cf}}, & \text{if } t \in [t_a, t_r], \\ \hat{y}_t^{\text{std}}, & \text{otherwise.} \end{cases} \quad (8)$$

The critic is then trained by minimizing the temporal-difference error to the corrected target:

$$\mathcal{L}_{\text{critic}} = \frac{1}{2} \sum_{i=1}^2 \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathcal{B}} \left[(Q_{\phi_i}(s_t, a_t) - \hat{y}_t^{\text{corr}})^2 \right]. \quad (9)$$

3.4 Preference Auxiliary Policy Optimization

While the corrected critic suppresses value inflation on suboptimal segments, the intervention signal can also directly guide the policy distribution learning. For each intervention state s_{t_h} , we obtain a zero-cost preference pair $(a^{\text{human}}, a^\pi)$. A direct log-probability preference objective, as in discrete-action preference optimization [30], is ill-conditioned for continuous tanh-squashed Gaussian actors. As training converges and the policy variance shrinks, log-probability gaps become dominated by the $1/\sigma_d^2$ term, leading to unstable gradients. We therefore impose the preference in the bounded mean-action space and backpropagate only through the squashed mean $\bar{a}_\theta(s) = \tanh(\mu_\theta(s))$:

$$\mathcal{L}_{\text{pref}} = E_{(s, a^{\text{human}}, a^\pi) \sim \mathcal{B}_{\text{pref}}} \left[\left\| \tanh(\mu_\theta(s)) - a^{\text{human}} \right\|^2 + \frac{1}{\left\| \tanh(\mu_\theta(s)) - a^\pi \right\|^2 + \epsilon} \right], \quad (10)$$

where the first term aligns the actor with the corrective action, the second repels it from the intervention-triggering policy action, and ϵ prevents numerical singularity. The final actor objective combines the standard RLPD actor loss with the preference-calibration term:

$$\mathcal{L}_{\text{actor}} = \mathcal{L}_{\text{RLPD}} + \lambda_{\text{pref}} \mathcal{L}_{\text{pref}}, \quad (11)$$

where λ_{pref} controls the strength of actor-side preference calibration.

4 Experimental Results

We evaluate PACT on real-robot manipulation tasks through four questions. (1). Does PACT improve sample efficiency and performance (Section 4.2)? (2). Can the progress model reliably identify suboptimal segments before human interventions (Section 4.3)? (3). Can preference-aware

Table 1: Real-world Results. Comparison of the success rate, intervention rate, training time, and actor steps between PACT and Baselines across six real-world tasks.

Metric	Method	Press	Insertion	Pick	Pick & Place	Folding	Assembly	Average
Success Rate (%) $\uparrow$	RLIF	75	60	50	25	10	0	36.7
	HIL-SERL	85	73	67	55	30	10	53.3
	PACT	95	90	95	70	70	63	80.5
Intervention Rate (%) $\downarrow$	RLIF	37.4	36.7	38.9	63.8	52.4	59.6	48.1
	HIL-SERL	43.6	45.3	35.8	54.9	63.1	56.0	49.8
	PACT	19.1	32.5	26.1	36.6	40.9	47.2	33.7
Training Time (Minutes) $\downarrow$	RLIF	37.7	51.4	74.8	108.4	106.2	122.1	83.4
	HIL-SERL	23.2	51.8	75.0	110.8	118.2	142.6	86.9
	PACT	21.4	30.3	46.6	94.5	90.1	122.1	67.5
Actor Steps (K) $\downarrow$	RLIF	13.1	18.6	23.9	35.8	33.5	40.4	27.6
	HIL-SERL	7.7	17.6	22.1	38.0	38.3	44.1	28.0
	PACT	7.3	11.7	15.7	31.3	27.9	39.9	22.3

counterfactual correction mitigate Q-value overestimation during training (Section 4.4)? (4). How does each component contribute to the policy performance improvement (Section 4.5)?

4.1 Experiment Setup

Real Robot Task. We evaluate PACT on six real-world manipulation tasks with increasing difficulty: *Press*, *Insertion*, *Pick*, *Pick & Place*, *Folding* and *Assembly*. For each task, we collect 20 demonstration trajectories and store them in $\mathcal{D}_{demo}$ for progress model training and early-stage RL stabilization. All experiments are conducted on a *Galaxea AIX* 6-DoF arm.

Baselines. We adopt HIL-SERL [5] and RLIF [31] as the primary baselines. HIL-SERL incorporates human intervention trajectories into replay-based off-policy training with the entropy-regularized RLPD objective and represents the state-of-the-art HIL-RL framework for real-world manipulation. RLIF uses human intervention signals as rewards for off-policy reinforcement learning, penalizing transitions immediately preceding interventions without requiring a manually specified task reward. During training, the operator intervenes only when abnormal behavior occurs and takes over the rollout until completion, following the intervention principle used in HIL-SERL.

Evaluation Metrics. We evaluate each method using four metrics: final success rate (SR), training time, average human intervention rate (IR), and actor steps. Actor steps denote the number of robot–environment interaction steps. The human intervention rate is defined as the ratio between the number of intervention steps and the total robot interaction steps.

4.2 Comparisons with the State-of-the-Art

Table 1 reports final quantitative results for PACT and baselines, while Fig. 3 shows training dynamics for PACT and HIL-SERL in intervention and success rates across six tasks. PACT consistently improves final performance and sample efficiency, increasing the average success rate from 53.3% to 80.5% and reducing the average intervention rate from 49.8% to 33.7% compared with HIL-SERL. It also outperforms RLIF, which achieves an average success rate of 36.7% and an intervention rate of 48.1%. The improvement is especially pronounced on *Assembly*, the most complex task, where PACT reaches a 63% success rate in 122 minutes, compared with only 10% for HIL-SERL over 140 minutes. PACT also shortens average training time to 67.5 minutes, compared with 86.9 minutes for HIL-SERL and 83.4 minutes for RLIF. PACT also uses fewer robot–environment interaction steps on average: 22.3K, versus 28.0K for HIL-SERL and 27.6K for RLIF. The learning curves in Fig. 3 further show that PACT reduces human intervention more rapidly while achieving faster success-rate growth, suggesting that preference-aware segment-level correction improves policy learning rather than merely relying on additional human recovery.

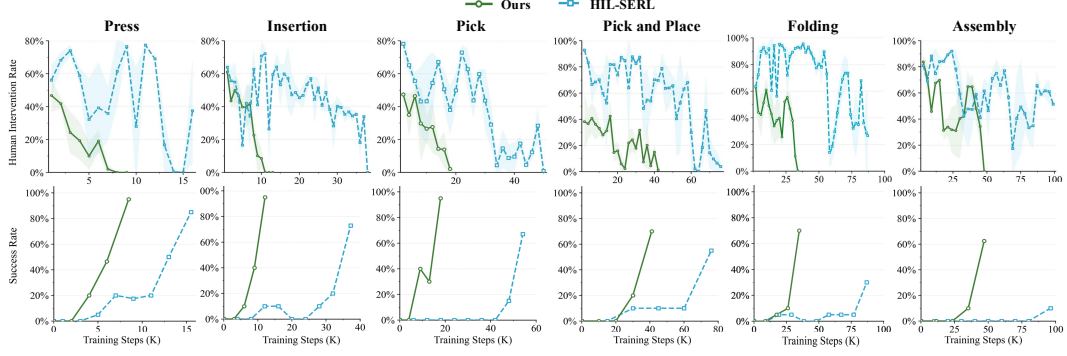


Figure 3: Real-robot training dynamics of PACT and HIL-SERL across six manipulation tasks.

4.3 Progress Model Analysis

Qualitative Results. We first examine whether the progress model localizes behaviorally meaningful suboptimal segments for subsequent credit correction. Fig. 4 visualizes representative real-robot rollouts, where the end-effector trajectories are plotted in 3D space and the segments detected by the progress model are highlighted in red. These segments align with typical failure patterns, such as hesitation in *Insertion*, wrong grasp in *Pick*, and directional deviation in *Pick & Place*, suggesting that the identified results are closely associated with degraded execution, providing a basis for targeting segment-level credit correction at the detected suboptimal parts of intervention-containing successful trajectories.

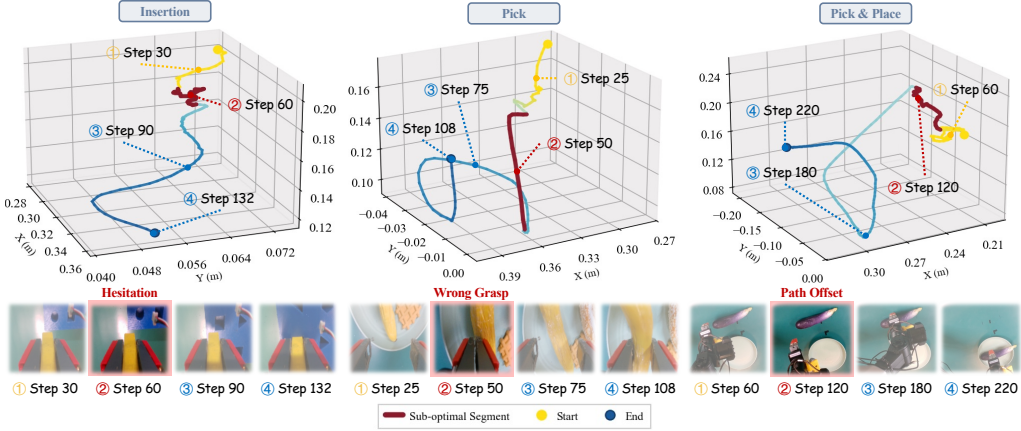


Figure 4: Progress-based suboptimal segment localization. The red segments denote identified suboptimal portions, which correspond to wrong behavior in RL rollouts.

Table 2: Trajectory quality comparison between normal and identified suboptimal segments. η denotes path efficiency, and $\Delta\theta$ denotes the direction deviation to the nearest demonstration segment.

Metric	Segment	Press	Insertion	Pick	Pick & Place	Folding	Assembly
Path Efficiency $\eta \uparrow$	Normal	.530	.577	.551	.537	.546	.495
	Suboptimal	.521	.471	.490	.432	.487	.473
Direction Deviation $\Delta\theta \downarrow$	Normal	47.1°	41.7°	40.4°	54.8°	43.0°	38.8°
	Suboptimal	67.4°	67.4°	68.3°	83.1°	72.2°	63.7°

Quantitative Evaluation. Since the progress model is trained without external supervised labels, we assess its localization results using two post-hoc, model-independent kinematic metrics: path

Table 3: Average critic bias $Q(s, a) - G_t$ on detected suboptimal segments. Positive values indicate overestimation relative to the Monte Carlo return.

Method	Press	Insertion	Pick	Pick & Place	Folding	Assembly
HIL-SERL	+0.029	+0.114	+0.019	+0.054	+0.082	+0.060
PACT	-0.042	-0.061	-0.107	-0.146	-0.113	-0.152

efficiency η and direction deviation $\Delta\theta$. For a trajectory segment $[t_{\text{start}}, t_{\text{end}}]$, they are defined as:

$$\eta = \frac{\|\text{pos}(t_{\text{end}}) - \text{pos}(t_{\text{start}})\|}{\sum_{t=t_{\text{start}}}^{t_{\text{end}}-1} \|\text{pos}(t+1) - \text{pos}(t)\|}, \quad \Delta\theta = \arccos\left(\frac{\vec{d}_{\text{seg}} \cdot \vec{d}_{\text{demo}}}{\|\vec{d}_{\text{seg}}\| \|\vec{d}_{\text{demo}}\|}\right). \quad (12)$$

where $\vec{d}_{\text{seg}} = \text{pos}(t_{\text{end}}) - \text{pos}(t_{\text{start}})$, $\vec{d}_{\text{demo}} = \text{pos}_{\text{demo}}(t_{\text{match}} + N) - \text{pos}_{\text{demo}}(t_{\text{match}})$, and $N = t_{\text{end}} - t_{\text{start}}$. Here, t_{match} is the timestep in $\mathcal{D}_{\text{demo}}$ whose end-effector position is nearest to $\text{pos}(t_{\text{start}})$. These metrics capture complementary and task-agnostic aspects of motion quality, where η measures motion redundancy by comparing net displacement with accumulated path length and $\Delta\theta$ measures execution drift as the angular mismatch between the segment direction and the nearest matched demonstration direction. As shown in Table 2, identified suboptimal segments consistently exhibit lower path efficiency and larger direction deviation, providing a descriptive characterization of their motion patterns.

4.4 Q-value Correction Analysis

We further evaluate whether preference-aware counterfactual correction suppresses Q-value overestimation during training. Since true action values are unavailable in real-robot experiments, we use the Monte Carlo return G_t only as a diagnostic reference and compare the learned critics of HIL-SERL and PACT on intervention-containing successful episodes.

Fig. 5(a) compares the target Q-value evolution during training on the *Insertion* task. HIL-SERL produces rapidly increasing target values, suggesting that uniform TD backup propagates terminal rewards to suboptimal pre-intervention segments and leads to value inflation. In contrast, PACT keeps the value estimates more conservative through preference-aware correction. Fig. 5(b) shows a representative case. Within the detected correction segment, HIL-SERL assigns values higher than the MC reference, whereas PACT substantially lowers the estimated values. Fig. 5(c) further aggregates the normalized critic bias $Q(s, a) - G_t$ over 20 episodes, where HIL-SERL exhibits consistently positive bias while PACT largely suppresses this overestimation. Table 3 confirms this trend across all five tasks.

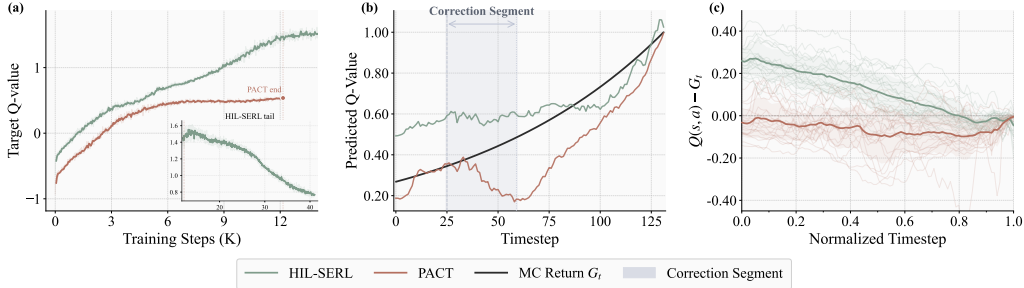


Figure 5: Q-value correction analysis. (a) Training stage Q-value evolution. (b) Single-trajectory value profile. (c) Normalized critic bias over intervention-containing successful episodes.

4.5 Ablation Study

Component Ablation. We conduct component ablations on *Insertion* and *Assembly*. Here, C1 denotes the critic-side correction module, including progress-based segment localization with Q-value correction, and C2 denotes the actor-side preference loss. As shown in Table 4(a), removing

Table 4: Ablation studies. (a) Component ablations on *Insertion* and *Assembly*. (b) Comparison of segment localization strategies on *Insertion*. Training durations are approximately 30 minutes for *Insertion* and 122 minutes for *Assembly*. SR and IR are reported in percentages.

(a) Component ablation					(b) Corrected interval ablation		
Setting	<i>Insertion</i>		<i>Assembly</i>		Setting	<i>Insertion</i>	
	SR $\uparrow$	IR $\downarrow$	SR $\uparrow$	IR $\downarrow$		SR $\uparrow$	IR $\downarrow$
w/o. C1	60	41.3	19	59.1	Fixed $L = 3$	85	29.9
w/o. C2	80	45.0	44	55.4	Fixed $L = 10$	75	31.1
PACT	90	32.5	63	47.2	PACT	90	32.5

C1 causes the largest success-rate drop on both tasks, while removing either component increases the intervention rate. These results support the complementary contributions of critic-side correction and actor-side preference learning. Table 4(b) further compares progress-based localization with fixed correction lengths on *Insertion*. Although fixed lengths yield slightly lower intervention rates, PACT achieves a higher success rate, supporting the benefit of progress-based localization for RL performance.

Parameter Sensitivity Analysis. We evaluate $\beta \in \{1, 3, 5\}$ and $\lambda_{\text{pref}} \in \{0.1, 0.2, 0.3\}$ on *Insertion*, using $\beta = 3$ and $\lambda_{\text{pref}} = 0.2$ as the default setting. Table 5 reports the default performance and mean absolute changes from this setting. For localization, we also examine $\delta_{\text{reg}} \in [0.025, 0.065]$ and $W \in [1, 5]$; small values of W can miss gradual declines in predicted progress. The same default configuration is used across tasks without task-specific tuning.

Table 5: Sensitivity on Insertion. The default row reports SR and IR (%), * denotes mean absolute differences from the default

Setting	Range	SR	IR
β^*	[1, 5]	7.5	4.2
λ_{pref}^*	[0.1, 0.3]	5.0	3.9
Default	—	90.0	32.5

5 Conclusion

In this paper, we propose a Preference-Calibrated HIL-RL method that treats intervention trajectories as structured heterogeneous data. By localizing suboptimal segments, correcting inflated Bellman targets with intervention-induced preferences, and aligning the actor toward human preference actions, PACT improves success rates, reduces human intervention, and accelerates training across six real-robot manipulation tasks. Further analyses show that the progress model, critic correction, and actor-side preference objective contribute complementary benefits for sample-efficient real-world robot learning.

6 Limitation

While PACT substantially improves sample efficiency in real-world HIL-RL, it does not fully resolve credit assignment under all manipulation conditions. To keep the method lightweight and annotation-free for real-robot training, the progress model relies on limited demonstration-based self-supervision and may be affected by insufficient demonstration coverage, non-monotonic trajectories or repetitive execution patterns. Since the subsequent critic correction depends on the localized suboptimal segments, such localization errors may further propagate to critic learning. Moreover, the proposed counterfactual advantage provides a directional preference-based correction rather than a precise per-step value target, making it more suitable for suppressing systematic overestimation than for fine-grained value calibration. Generalization to broader out-of-distribution scene reconfigurations remains untested. Future work could incorporate semantic-rich progress models, VLM-assisted progress reasoning, and pretrained progress representations to improve robustness and enable more accurate credit reassignment in complex long-horizon real-robot tasks.

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